BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 050 · Lesson 050 · Queue<T>: FIFO and Service Order

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is stack Pop order?

Q2. What is Peek?

Q3. What does Enqueue do?

Q4. 021

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

FIFO — First In First Out.

Enqueue — Add at queue rear.

Dequeue — Remove from queue front.

front — Next item to be served.

New learning

Observe — A queue is first-in, first-out (FIFO). Enqueue joins the rear; Dequeue removes from the front.

Explain — Queues are suitable for ordered service requests such as print jobs or breadth-first exploration, if the queueing policy is FIFO.

Apply — Peek inspects the front without removing. An empty Queue<T> throws on Dequeue/Peek unless the code guards against it.

Worked example

Trace a ticket queue with Jo then Kai; dequeue one.

1. Trace and justify — Jo is served first; Kai remains at the front.

We Do · Practise with guidance

A different queue receives jobs 4, 7, 2. Dequeue twice, enqueue 9. What remains?

```
var q=new Queue<int>();
q.Enqueue(4);q.Enqueue(7);q.Enqueue(2);
q.Dequeue();q.Dequeue();q.Enqueue(9);
```

Trace and justify

Dequeue returns 4 then 7. Remaining front-to-back [2,9].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Queue A,B,C; Dequeue once, then Peek. State removed and front value. [6]

Question 2

Write C# code to enqueue 2,4,6 into Queue<int>, then safely dequeue one. [6]

Question 3

Compare Stack<T> and Queue<T> for DFS versus BFS on a graph, explaining removal order. [6]

Exit Ticket · Answer before checking

A queue's Dequeue removes the newest item first.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Last-in-first-out.

Q2. Read an end item without removal.

Q3. Adds to rear of queue.

Q4. Enqueue A,B,C. Dequeue once then Enqueue D. Give queue front-to-back.

You Do pages · Independent questions

Q1. Dequeue A [2]; Peek B [2]; remaining front-to-back B,C [2].

Q2. var q=new Queue<int>(); [1]; Enqueue 2,4,6 [3]; if(q.Count>0) [1] q.Dequeue() [1].

Q3. DFS can use stack LIFO [2] to explore most recently discovered branch [1]; BFS uses queue FIFO [2] to process older frontiers first [1].

Exit Ticket · Model answer

False. FIFO removes the oldest queued item.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: